

Lecture 4

numerical differentiation

first derivative

Definition of the first derivative

Let $f(x)$ be right- and left-differentiable at some point x . Then, with $h > 0$,

$$f'_R(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (\text{right first derivative})$$

but also

$$f'_L(x) = \lim_{h \rightarrow 0} \frac{f(x) - f(x-h)}{h} \quad (\text{left first derivative})$$

If $f(x)$ is differentiable at x , then

$$f'_R(x) = f'_L(x) = f'(x) \quad (\text{first derivative})$$

Forward and backward differences

With $h > 0$:

$$Q_F(h) = \frac{f(x+h) - f(x)}{h} \quad (\text{forward difference})$$

$$Q_B(h) = \frac{f(x) - f(x-h)}{h} \quad (\text{backward difference})$$

Approximation of derivative by forward difference

The idea of the *finite-difference approximation*:

$$f'(x) \approx Q_F(h)$$

To be precise:

$$f'(x) = Q_F(h) + R_F(h)$$

The *truncation error* (afbreekfout) of the forward-difference approximation of $f'(x)$ is defined as

$$R_F(h) = f'(x) - Q_F(h) = f'(x) - \frac{f(x+h) - f(x)}{h}$$

How big is this error?

Forward-difference truncation error

Use the Taylor series to express $f(x+h)$ via $f(x)$ and its derivatives:

$$f(x+h) = f(x) + f'(x)h + \frac{f''(\xi)}{2}h^2, \quad \xi \in (x, x+h)$$

The truncation error:

$$\begin{aligned} R_F(h) &= f'(x) - Q_F(h) = f'(x) - \frac{f(x+h) - f(x)}{h} \\ &= f'(x) - \frac{1}{h} \left(f(x) + f'(x)h + \frac{f''(\xi)}{2}h^2 - f(x) \right) \\ &= -\frac{f''(\xi)}{2}h, \quad \xi \in (x, x+h) \end{aligned}$$

Order of the forward-difference approximation

$$Q_F(h) = \frac{f(x+h) - f(x)}{h},$$

$$R_F(h) = -\frac{f''(\xi)}{2}h = \mathcal{O}(h)$$

$$f'(x) = Q_F(h) + \mathcal{O}(h)$$

The forward-difference formula $Q_F(h)$ is an $\mathcal{O}(h)$ (order- h) approximation of $f'(x)$. The truncation error $R_F(h)$ goes to zero as fast as h does, i.e., linearly.

What does it mean: if you make h two times smaller, then the error will also become two times smaller.

Is it possible to make a better approximation? With the error that goes to zero faster than h ?

Central-difference approximation

$$Q_C(h) = \frac{f(x+h) - f(x-h)}{2h} \quad (\text{central difference})$$

$$f'(x) = Q_C(h) + R_C(h)$$

What is the truncation error $R_C(h)$?

Central-difference truncation error

Use the Taylor series for $f(x+h)$ and $f(x-h)$:

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + \frac{f'''(\xi_+)}{6}h^3, \quad \xi_+ \in (x, x+h)$$

$$f(x-h) = f(x) - f'(x)h + \frac{f''(x)}{2}h^2 - \frac{f'''(\xi_-)}{6}h^3, \quad \xi_- \in (x-h, x)$$

The truncation error:

$$\begin{aligned} R_C(h) &= f'(x) - Q_C(h) = f'(x) - \frac{f(x+h) - f(x-h)}{2h} \\ &= f'(x) - \frac{1}{2h} \left(f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + \frac{f'''(\xi_+)}{6}h^3 \right. \\ &\quad \left. - f(x) + f'(x)h - \frac{f''(x)}{2}h^2 + \frac{f'''(\xi_-)}{6}h^3 \right) \end{aligned}$$

Central-difference truncation error

$$R_C(h) = -\frac{f'''(\xi_+) + f'''(\xi_-)}{12}h^2, \quad \xi_+ \in (x, x+h), \quad \xi_- \in (x-h, x)$$

Assuming that $f(x) \in C^3(x-h, x+h)$, we can make this formula simpler. Using the *Intermediate Value Theorem* here! The number $(f'''(\xi_+) + f'''(\xi_-))/2$ is the average of $f'''(\xi_+)$ and $f'''(\xi_-)$. Hence, it is *between* $f'''(\xi_+)$ and $f'''(\xi_-)$. Hence, there is $\xi \in (\xi_-, \xi_+)$ such that

$$f'''(\xi) = \frac{f'''(\xi_+) + f'''(\xi_-)}{2}$$

Thus,

$$R_C(h) = -\frac{f'''(\xi)}{6}h^2, \quad \xi \in (x-h, x+h)$$

Order of the central-difference approximation

$$Q_C(h) = \frac{f(x+h) - f(x-h)}{2h},$$

$$R_C(h) = -\frac{f'''(\xi)}{6}h^2 = \mathcal{O}(h^2)$$

$$f'(x) = Q_C(h) + \mathcal{O}(h^2)$$

The central-difference formula $Q_C(h)$ is an $\mathcal{O}(h^2)$ approximation of $f'(x)$. The truncation error $R_C(h)$ goes to zero as fast as h^2 does, i.e., quadratically, which is faster than h .

What does it mean: if you make h two times smaller, then the error will become four times smaller.

Is it possible to make an even better approximation? With the error that goes to zero faster than h^2 ? How many points would you need?

Round-off error of the central-difference formula

On the computer we also have the round-off errors. Let $\tilde{y} = \text{fl}(y)$.

$$\tilde{Q}_C = \frac{\tilde{f}(x+h) - \tilde{f}(x-h)}{2\tilde{h}}$$

Using the expressions:

$$\tilde{f}(x+h) = f(x+h)(1 + \epsilon_+),$$

$$\tilde{f}(x-h) = f(x-h)(1 + \epsilon_-),$$

$$\tilde{h} = h(1 + \epsilon_h),$$

we get

$$\tilde{Q}_C = \left(\frac{f(x+h) - f(x-h)}{2h} + \frac{f(x+h)}{2} \frac{\epsilon_+}{h} - \frac{f(x-h)}{2} \frac{\epsilon_-}{h} \right) \frac{1}{1 + \epsilon_h}$$

Round-off error of the central-difference formula

$$\tilde{Q}_C = \left(Q_C + \frac{f(x+h)}{2} \frac{\epsilon_+}{h} - \frac{f(x-h)}{2} \frac{\epsilon_-}{h} \right) \frac{1}{1 + \epsilon_h}$$

To simplify this we use the Taylor series for $(1+x)^{-1}$ around $x=0$:

$$\frac{1}{1 + \epsilon_h} = 1 - \epsilon_h + \mathcal{O}(\epsilon_h^2)$$

$$\begin{aligned} \tilde{Q}_C = Q_C(1 - \epsilon_h + \mathcal{O}(\epsilon_h^2)) &+ \frac{f(x+h)}{2} \frac{\epsilon_+(1 - \epsilon_h + \mathcal{O}(\epsilon_h^2))}{h} \\ &- \frac{f(x-h)}{2} \frac{\epsilon_-(1 - \epsilon_h + \mathcal{O}(\epsilon_h^2))}{h} \end{aligned}$$

Round-off error depends on h

$$\begin{aligned}\tilde{Q}_C - Q_C &= -Q_C(-\epsilon_h + \mathcal{O}(\epsilon_h^2)) + \frac{f(x+h)}{2} \frac{\epsilon_+(1 - \epsilon_h + \mathcal{O}(\epsilon_h^2))}{h} \\ &\quad - \frac{f(x-h)}{2} \frac{\epsilon_-(1 - \epsilon_h + \mathcal{O}(\epsilon_h^2))}{h} \\ &= \left(-hQ_C(-\epsilon_h + \mathcal{O}(\epsilon_h^2)) + \frac{f(x+h)}{2} \epsilon_+(1 - \epsilon_h + \mathcal{O}(\epsilon_h^2)) \right. \\ &\quad \left. - \frac{f(x-h)}{2} \epsilon_-(1 - \epsilon_h + \mathcal{O}(\epsilon_h^2)) \right) \frac{1}{h},\end{aligned}$$

$$|\tilde{Q}_C - Q_C| = \frac{\hat{\epsilon}}{h},$$

where

$$\hat{\epsilon} = \left| hQ_C\epsilon_h + \frac{f(x+h)}{2}\epsilon_+ - \frac{f(x-h)}{2}\epsilon_- + \mathcal{O}(\epsilon_m^2) \right|$$

Total error of the central-difference approximation

In exact arithmetic the central-difference formula $Q_C(h)$ introduces only the truncation error. In finite-precision floating-point arithmetic, we have both **truncation** and **round-off** errors. The total error is bounded by:

$$\begin{aligned}|E_C(h)| &= |f'(x) - \tilde{Q}_C(h)| \leq |f'(x) - Q_C(h)| + |\tilde{Q}_C(h) - Q_C(h)| \\ &= \frac{|f'''(\xi)|}{6} h^2 + \frac{\hat{\epsilon}}{h} \leq \frac{M}{6} h^2 + \frac{\hat{\epsilon}}{h},\end{aligned}$$

where we used the *triangular inequality* and assumed that

$$|f'''(\xi)| \leq M \quad \text{for all } \xi \in (x - h, x + h)$$

Minimal useful step size

From the formula

$$|E_C(h)| \leq \frac{M}{6} h^2 + \frac{\hat{\epsilon}}{h} = \phi(h),$$

it follows that the upper bound $\phi(h)$ on the total error $|E_C(h)|$ has a minimal value $\phi(h_{\text{opt}})$ and it, probably, does not make sense to decrease h below h_{opt} .

$$\phi'(h) = \frac{M}{3} h - \frac{\hat{\epsilon}}{h^2} = 0,$$

$$h_{\text{opt}} = \left(\frac{3\hat{\epsilon}}{M} \right)^{1/3}$$

$$|E_C(h_{\text{opt}})| \leq \left(\frac{9M\hat{\epsilon}^2}{8} \right)^{1/3}$$

general approach to difference formulas

- It is intuitively clear that the difference formulas approximating the first derivative have the form

$$Q(h) = \frac{\alpha_1 f(x_1) + \alpha_2 f(x_2) + \cdots + \alpha_n f(x_n)}{h}$$

- We only need to choose x_i and α_i to maximize the order of the truncation error.

Uniform grid

Consider a uniform grid with step h in the neighborhood of the point x :

$$\dots, x - 3h, x - 2h, x - h, x, x + h, x + 2h, x + 3h, \dots$$

The corresponding function values (data points) are:

$$\dots, f(x - 3h), f(x - 2h), f(x - h), f(x), f(x + h), f(x + 2h), f(x + 3h), \dots$$

With three data points one has various possibilities, e.g.:

$$Q(h) = \frac{\alpha_{-1}f(x - h) + \alpha_0f(x) + \alpha_1f(x + h)}{h},$$

$$Q(h) = \frac{\alpha_{-2}f(x - 2h) + \alpha_{-1}f(x - h) + \alpha_0f(x)}{h},$$

$$Q(h) = \frac{\alpha_0f(x) + \alpha_1f(x + h) + \alpha_2f(x + 2h)}{h}, \text{ etc.}$$

where α_i still need to be determined.

Central-difference formula again

Let's see how this works with the central-difference choice of data points:

$$Q(h) = \frac{\alpha_{-1}f(x-h) + \alpha_0f(x) + \alpha_1f(x+h)}{h},$$

Using the Taylor series for all shifted data points, keeping as many terms as there are data points:

$$f(x-h) = f(x) - f'(x)h + \frac{f''(x)}{2}h^2 + \mathcal{O}(h^3)$$

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + \mathcal{O}(h^3)$$

Central-difference formula again

Substituting these we get:

$$\begin{aligned} Q(h) = & \frac{1}{h} \left[\alpha_{-1} \left(f(x) - f'(x)h + \frac{f''(x)}{2}h^2 + \mathcal{O}(h^3) \right) \right. \\ & + \alpha_0 f(x) \\ & \left. + \alpha_1 \left(f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + \mathcal{O}(h^3) \right) \right] \end{aligned}$$

Now, assemble coefficients of all derivatives separately:

$$\begin{aligned} Q(h) = & \left(\frac{1}{h}\alpha_{-1} + \frac{1}{h}\alpha_0 + \frac{1}{h}\alpha_1 \right) f(x) \\ & + (-\alpha_{-1} + \alpha_1) f'(x) \\ & + \left(\frac{h}{2}\alpha_{-1} + \frac{h}{2}\alpha_1 \right) f''(x) + \mathcal{O}(h^2) \end{aligned}$$

Central-difference formula again

We want the difference formula $Q(h)$ to approximate $f'(x)$. Hence, we want

$$\begin{aligned}Q(h) &= \left(\frac{1}{h}\alpha_{-1} + \frac{1}{h}\alpha_0 + \frac{1}{h}\alpha_1 \right) f(x) \\&\quad + (-\alpha_{-1} + \alpha_1) f'(x) \\&\quad + \left(\frac{h}{2}\alpha_{-1} + \frac{h}{2}\alpha_1 \right) f''(x) + \mathcal{O}(h^2) \\&= \textcolor{blue}{(0)}f(x) + \textcolor{red}{(1)}f'(x) + \textcolor{blue}{(0)}f''(x) + \mathcal{O}(h^2)\end{aligned}$$

This leads to the algebraic equations:

$$\begin{aligned}\frac{1}{h}\alpha_{-1} + \frac{1}{h}\alpha_0 + \frac{1}{h}\alpha_1 &= 0, \\-\alpha_{-1} + \alpha_1 &= 1, \\\frac{h}{2}\alpha_{-1} + \frac{h}{2}\alpha_1 &= 0\end{aligned}$$

Notice, that we have three equations and three unknowns.

Central-difference formula again

Solving the linear system we get:

$$\alpha_{-1} = -\frac{1}{2},$$

$$\alpha_0 = 0,$$

$$\alpha_1 = \frac{1}{2},$$

which leads to the central-difference formula:

$$f'(x) = \frac{-f(x-h) + f(x+h)}{2h} + \mathcal{O}(h^2)$$

How many points to use for first derivative?

How many data points do you need to derive an $\mathcal{O}(h^3)$ difference formula for $f'(x)$? The extended $Q(h)$ formula will look like

$$Q(h) = (0)f(x) + (1)f'(x) + (0)f''(x) + (0)f'''(x) + \mathcal{O}(h^3)$$

We need to satisfy 4 equations. Hence, we need 4 α -unknowns. For example:

$$Q(h) = \frac{1}{h} (\alpha_{-1}f(x-h) + \alpha_0f(x) + \alpha_1f(x+h) + \alpha_2f(x+2h))$$

Thus, we need to use 4 grid points (4-point stencil).

Higher-accuracy forward difference formula

We are lucky to have the $\mathcal{O}(h^2)$ accuracy with the two-point central-difference formula. In general, we need three points to achieve this accuracy. Let's use three "forward" data points to derive an $\mathcal{O}(h^2)$ formula:

$$Q(h) = \frac{\alpha_0 f(x) + \alpha_1 f(x+h) + \alpha_2 f(x+2h)}{h},$$

We need these Taylor series:

$$\begin{aligned} f(x+h) &= f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + \mathcal{O}(h^3) \\ f(x+2h) &= f(x) + f'(x)2h + \frac{f''(x)}{2}(2h)^2 + \mathcal{O}(h^3) \end{aligned}$$

Higher-accuracy forward difference formula

Substituting and grouping the terms we get:

$$\begin{aligned} Q(h) &= \left(\frac{1}{h}\alpha_0 + \frac{1}{h}\alpha_1 + \frac{1}{h}\alpha_2 \right) f(x) \\ &\quad + (\alpha_1 + 2\alpha_2) f'(x) \\ &\quad + \left(\frac{h}{2}\alpha_1 + 2h\alpha_2 \right) f''(x) + \mathcal{O}(h^2) \end{aligned}$$

The requirement $Q(h) = (0)f(x) + (1)f'(x) + (0)f''(x) + \mathcal{O}(h^2)$ leads to the algebraic equations:

$$\begin{aligned} \frac{1}{h}\alpha_0 + \frac{1}{h}\alpha_1 + \frac{1}{h}\alpha_2 &= 0, \\ \alpha_1 + 2\alpha_2 &= 1, \\ \frac{h}{2}\alpha_1 + 2h\alpha_2 &= 0. \end{aligned}$$

with the solution: $\alpha_0 = -3/2$, $\alpha_1 = 2$, $\alpha_2 = -1/2$.

Higher-accuracy forward difference formula

Thus, we obtain an $\mathcal{O}(h^2)$ three-point forward-difference formula:

$$Q(h) = \frac{-3f(x) + 4f(x+h) - f(x+2h)}{2h} = f'(x) + \mathcal{O}(h^2)$$

approximating the second derivative

Deriving a difference formula for $f''(x)$

The general rules:

- In the formula for the n -th derivative, we divide the Taylor series by the term h^n , e.g., for $f''(x)$, we divide by h^2 .
- To eliminate all derivatives before $f^{(n)}(x)$ in the Taylor series, we need at least $n + 1$ data points, e.g., at least three points for $f''(x)$.
- To obtain the formula of the $\mathcal{O}(h^p)$, $p \geq 1$, truncation accuracy for $f^{(n)}(x)$, in general, we need $n + p$ grid points, e.g., $2 + 2 = 4$ points to obtain an $\mathcal{O}(h^2)$ formula for $f''(x)$.

For example, using the four-point formula for $f''(x)$

$$Q(h) = \frac{1}{h^2} (\alpha_0 f(x_0) + \alpha_1 f(x_1) + \alpha_2 f(x_2) + \alpha_3 f(x_3))$$

we will need to enforce the following four conditions

$$Q(h) = (0)f(x) + (0)f'(x) + (1)f''(x) + (0)f'''(x) + \mathcal{O}(h^2)$$

Central-difference formula for $f''(x)$

With the symmetric allocation of grid points, we often need less points to achieve the desired accuracy, e.g. three points, instead of four to achieve $\mathcal{O}(h^2)$:

$$Q(h) = \frac{\alpha_{-1}f(x-h) + \alpha_0f(x) + \alpha_1f(x+h)}{h^2},$$

we need the Taylor series:

$$f(x-h) = f(x) - f'(x)h + \frac{f''(x)}{2}h^2 - \frac{f'''(x)}{6}h^3 + \mathcal{O}(h^4)$$

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + \frac{f'''(x)}{6}h^3 + \mathcal{O}(h^4)$$

Deriving a difference formula for $f''(x)$

Substituting and collecting the terms we get:

$$\begin{aligned} Q(h) = & \left(\frac{1}{h^2} \alpha_{-1} + \frac{1}{h^2} \alpha_0 + \frac{1}{h^2} \alpha_1 \right) f(x) \\ & + \left(-\frac{1}{h} \alpha_{-1} + \frac{1}{h} \alpha_1 \right) f'(x) \\ & + \left(\frac{1}{2} \alpha_{-1} + \frac{1}{2} \alpha_1 \right) f''(x) \\ & + \left(-\frac{h}{6} \alpha_{-1} + \frac{h}{6} \alpha_1 \right) f'''(x) + \mathcal{O}(h^2) \end{aligned}$$

Now we require that

$$Q(h) = (0)f(x) + (0)f'(x) + (1)f''(x) + (0)f'''(x) + \mathcal{O}(h^2) = f''(x) + \mathcal{O}(h^2)$$

Deriving a difference formula for $f''(x)$

This leads to the algebraic equations:

$$\begin{aligned}\frac{1}{h^2}\alpha_{-1} + \frac{1}{h^2}\alpha_0 + \frac{1}{h^2}\alpha_1 &= 0, \\ -\frac{1}{h}\alpha_{-1} + \frac{1}{h}\alpha_1 &= 0, \\ \frac{1}{2}\alpha_{-1} + \frac{1}{2}\alpha_1 &= 1, \\ -\frac{h}{6}\alpha_{-1} + \frac{h}{6}\alpha_1 &= 0,\end{aligned}$$

which are solved by:

$$\begin{aligned}\alpha_{-1} &= 1, \\ \alpha_0 &= -2, \\ \alpha_1 &= 1.\end{aligned}$$

From the point of view of linear algebra, we are lucky to have a solution.
Why?

A difference formula for $f''(x)$

Hence,

$$Q(h) = \frac{f(x-h) - 2f(x) + f(x+h)}{h^2} = f''(x) + \mathcal{O}(h^2),$$

or

$$f''(x) = \frac{f(x-h) - 2f(x) + f(x+h)}{h^2} + \mathcal{O}(h^2),$$

and we have obtained an $\mathcal{O}(h^2)$ central-difference formula for $f''(x)$.

Determining the truncation error of a given formula

After this derivation we only know that the difference formula is $\mathcal{O}(h^2)$. To determine the exact form of the truncation error we need to compute:

$$f''(x) - Q(h) = f''(x) - \frac{1}{h^2}(f(x-h) - 2f(x) + f(x+h))$$

Use the Taylor series for the shifted data points with explicit remainders:

$$f(x-h) = f(x) - f'(x)h + \frac{f''(x)}{2}h^2 - \frac{f'''(x)}{6}h^3 + \frac{f^{(4)}(\xi_-)}{24}h^4$$

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + \frac{f'''(x)}{6}h^3 + \frac{f^{(4)}(\xi_+)}{24}h^4$$

Determining the truncation error of a given formula

Substitution gives

$$\begin{aligned}f''(x) - Q(h) &= f''(x) - \frac{1}{h^2} \left(f(x) - f'(x)h + \frac{f''(x)}{2}h^2 - \frac{f'''(x)}{6}h^3 + \frac{f^{(4)}(\xi_-)}{24}h^4 \right. \\&\quad \left. - 2f(x) + f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + \frac{f'''(x)}{6}h^3 + \frac{f^{(4)}(\xi_+)}{24}h^4 \right) \\&= -\frac{f^{(4)}(\xi_-) + f^{(4)}(\xi_+)}{2} \frac{h^2}{12} = -\frac{f^{(4)}(\xi)}{12} h^2, \quad \xi \in (x-h, x+h),\end{aligned}$$

where we have used the Intermediate Value Theorem.

Richardson's extrapolation method

General expression for the truncation error

We have seen that the truncation error of finite-difference formulas for the derivatives had the form of:

$$M - Q(h) = c_p(h)h^p$$

For example, in the forward-difference approximation of $f'(x)$ we have: $M = f'(x)$, $p = 1$, $c_1(h) = -f''(\xi)/2$, and $\xi \in (x, x + h)$. Note, ξ depends on h .

If in the formula for truncation error we use a constant c_p that does not depend on h , then the formula can be written as

$$M - Q(h) = c_p h^p + \mathcal{O}(h^{p+1}),$$

where we assume the existence of the $p + 1$ derivative of the function that is being differentiated.

Idea of Richardson's extrapolation

The idea of Richardson's extrapolation is to use several estimates $Q(h)$ with different h 's and determine the unknown parameters c_p and p of the error term numerically. This gives a good, $\mathcal{O}(h^{p+1})$ estimate of the truncation error and even allows to improve (extrapolate) the formula $M \approx Q(h)$, which is $\mathcal{O}(h^p)$, to the formula $M \approx Q(h) + c_p h^p$, which is $\mathcal{O}(h^{p+1})$.

Determining the parameters of the Richardson extrapolation

Use the formula $Q(h)$ to produce three estimates of the unknown M , e.g.: $Q(h)$, $Q(2h)$, $Q(4h)$. Neglecting the $\mathcal{O}(h^{p+1})$ term we obtain three equations:

$$\begin{aligned}M - Q(h) &= c_p h^p, \\M - Q(2h) &= c_p (2h)^p = c_p 2^p h^p, \\M - Q(4h) &= c_p (4h)^p = c_p 2^{2p} h^p,\end{aligned}$$

Leads to:

$$\begin{aligned}Q(2h) - Q(4h) &= c_p h^p (2^p - 1) 2^p, \\Q(h) - Q(2h) &= c_p h^p (2^p - 1),\end{aligned}$$

and

$$\frac{Q(2h) - Q(4h)}{Q(h) - Q(2h)} = 2^p$$

Determining the parameters of the Richardson extrapolation

From

$$Q(h) - Q(2h) = c_p h^p (2^p - 1),$$

we obtain

$$c_p h^p = \frac{Q(h) - Q(2h)}{2^p - 1} = \frac{Q(h) - Q(2h)}{\frac{Q(2h) - Q(4h)}{Q(h) - Q(2h)} - 1} = \frac{(Q(h) - Q(2h))^2}{2Q(2h) - Q(4h) - Q(h)}$$

Thus,

$$M - Q(h) = \frac{Q(h) - Q(2h)}{2^p - 1} + \mathcal{O}(h^{p+1}),$$

which is useful if we happen to know p . Alternatively,

$$M - Q(h) = \frac{(Q(h) - Q(2h))^2}{2Q(2h) - Q(4h) - Q(h)} + \mathcal{O}(h^{p+1})$$

Extrapolating $Q(h)$ to a higher order

Extrapolation of $Q(h)$ can be obtained as:

$$M = Q(h) + \frac{Q(h) - Q(2h)}{2^p - 1} + \mathcal{O}(h^{p+1}),$$

Consider the forward-difference formula for $f'(x)$:

$$f'(x) = Q(h) + \mathcal{O}(h) = \frac{f(x+h) - f(x)}{h} + \mathcal{O}(h)$$

From $\mathcal{O}(h)$, we know that $p = 1$ for this $Q(h)$. Extrapolating:

$$\begin{aligned} f'(x) &= Q(h) + \frac{Q(h) - Q(2h)}{2^1 - 1} + \mathcal{O}(h^{1+1}) \\ &= \frac{f(x+h) - f(x)}{h} + \frac{f(x+h) - f(x)}{h} - \frac{f(x+2h) - f(x)}{2h} + \mathcal{O}(h^2) \\ &= \frac{-3f(x) + 4f(x+h) - f(x+2h)}{2h} + \mathcal{O}(h^2). \end{aligned}$$

Previously, we derived this $\mathcal{O}(h^2)$ formula via a different approach.

Questions

- Compute the truncation error $R_B(h)$ of the backward-difference approximation $Q_B(h)$ of the first derivative $f'(x)$.
- Compute the minimal useful step size h_{opt} and the optimal upper bound on the total error $|E_B(h_{\text{opt}})|$ for the backward-difference approximation $\tilde{Q}_B(h)$ in finite precision floating-point arithmetic.
- Use general approach to derive a difference formula approximating $f'(x)$ to $\mathcal{O}(h^3)$.
- Let $f(x) = \sin(x)$. Compute $f'(1)$ using the central-difference formula with $h = 0.1$. Use Richardson's extrapolation to estimate the truncation error and to derive a better approximation for $f'(1)$.